

# Week 3 — z-Transform, ROC, Transfer Functions, and Pole–Zero Geometry

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DSP + ML • Senior / First-Year Graduate ECE • Oppenheim & Schaffer alignment

## Homework Problems

Show reasoning and ROC information. A rational z-transform without its ROC is not considered a complete answer when the time-domain sequence is being identified.

### Problem 1: Finite sequence z-transform

Let  $x[n] = 2\delta[n+1] - \delta[n] + 3\delta[n-2]$ . Find  $X(z)$ , state the ROC, and determine whether the DTFT exists.

### Problem 2: Right-sided exponential and convergence

For  $x[n] = (1.2)^n u[n]$ , derive  $X(z)$  from the geometric series, state the ROC, identify the pole, and determine whether its DTFT exists.

### Problem 3: Same algebra, different ROC

Find the inverse z-transform of  $X(z) = 1/(1 - 0.75z^{-1})$  for (a)  $|z| > 0.75$  and (b)  $|z| < 0.75$ . Explain in one sentence why the two answers can have the same rational expression.

### Problem 4: Partial fractions with two ROCs

Let  $X(z) = 1/[(1 - 0.5z^{-1})(1 - 0.8z^{-1})]$ . Use partial fractions and find  $x[n]$  for (a)  $|z| > 0.8$  and (b)  $0.5 < |z| < 0.8$ .

### Problem 5: Difference equation to $H(z)$

The system obeys  $y[n] - 1.4y[n-1] + 0.45y[n-2] = x[n] + x[n-1]$ . Find  $H(z)$ , its zeros and poles, and the ROC if the system is causal. Is the causal system BIBO stable?

### Problem 6: Causality and stability from ROC

A rational system has poles at  $z = 0.6$  and  $z = 1.1$ . For each possible rational ROC— $|z| > 1.1$ ,  $0.6 < |z| < 1.1$ , and  $|z| < 0.6$ —state whether the corresponding impulse response is causal, anti-causal/two-sided as appropriate, and BIBO stable.

### Problem 7: Geometric frequency-response reasoning

Consider  $H(z) = (1 + z^{-1})/(1 - 0.8z^{-1})$ . Without first computing a dense frequency response, compare  $|H(\exp(j\omega))|$  near  $\omega = 0$  and  $\omega = \pi$  using the zero at  $-1$  and pole at  $+0.8$ . What qualitative filter type do you expect?

### Problem 8: 60-Hz notch design

At  $f_s = 1000$  Hz, design the pole-zero locations of a second-order notch at 60 Hz using zeros on the unit circle and poles at radius  $r = 0.96$  with the same angles. Give  $\omega_0$ , the two zero locations, the two pole locations, and explain what increasing  $r$  to 0.99 would do.

### Problem 9: Graduate stretch: pole radius and time-domain decay

A causal system has a complex pole pair  $p = r \exp(\pm j\theta)$ . Explain why the corresponding impulse-response terms decay roughly as  $r^n$  and why moving  $r$  toward 1 simultaneously increases ringing duration and sharpens the frequency-domain resonance. Connect your argument to geometric distance from the unit circle.

## Homework Solutions

### Solution 1

Using the sifting definition,  $X(z) = 2z - 1 + 3z^{-2}$ . Because both positive and negative powers occur but only finitely many terms are present, the ROC is  $0 < |z| < \infty$ . The unit circle is included, so the DTFT exists.

### Solution 2

$X(z) = \sum_{n=0}^{\infty} (1.2z^{-1})^n = 1/(1 - 1.2z^{-1})$ , which converges for  $|1.2z^{-1}| < 1$ , hence  $|z| > 1.2$ . The pole is  $z = 1.2$ . The unit circle is not in the ROC, so the ordinary DTFT does not exist.

### Solution 3

(a)  $|z| > 0.75$  corresponds to the right-sided sequence  $x[n] = (0.75)^n u[n]$ . (b)  $|z| < 0.75$  corresponds to  $x[n] = -(0.75)^n u[-n-1]$ . The rational expression is the same because it represents different Laurent-series expansions that converge in different regions.

### Solution 4

Partial fractions give  $X(z) = -(5/3)/(1-0.5z^{-1}) + (8/3)/(1-0.8z^{-1})$ . (a) For  $|z| > 0.8$  both are right-sided:  $x[n] = -(5/3)(0.5)^n u[n] + (8/3)(0.8)^n u[n]$ . (b) For  $0.5 < |z| < 0.8$ , the 0.5 term is right-sided and the 0.8 term is left-sided:  $x[n] = -(5/3)(0.5)^n u[n] - (8/3)(0.8)^n u[-n-1]$ .

### Solution 5

$H(z) = (1+z^{-1})/(1-1.4z^{-1}+0.45z^{-2})$ . The zero is  $z = -1$ . The denominator factors as  $(1-0.5z^{-1})(1-0.9z^{-1})$ , so poles are 0.5 and 0.9. Causal ROC:  $|z| > 0.9$ . Since the unit circle lies in that ROC (equivalently both poles are strictly inside it), the causal system is BIBO stable.

### Solution 6

ROC  $|z| > 1.1$ : right-sided/causal, but not stable because the unit circle is excluded. ROC  $0.6 < |z| < 1.1$ : two-sided and stable because  $|z| = 1$  lies in the ROC. ROC  $|z| < 0.6$ : left-sided/anti-causal and not stable because the unit circle is excluded. This problem illustrates why pole locations alone do not specify stability without an ROC/support assumption.

### Solution 7

The zero at  $-1$  lies exactly at the unit-circle point for  $\omega = \pi$ , so  $|H(\exp(j\pi))| = 0$ . The pole at  $+0.8$  is close to the unit-circle point  $+1$  for  $\omega = 0$ , making the denominator distance small and the low-frequency gain large. Therefore the system has strong low-pass behavior with an exact zero at Nyquist.

### Solution 8

$\omega_0 = 2\pi(60/1000) = 0.37699$  rad/sample. Zeros are  $\exp(\pm j\omega_0) \approx 0.9298 \pm j0.3681$ . Poles are  $0.96\exp(\pm j\omega_0) \approx 0.8926 \pm j0.3534$ . Moving  $r$  from 0.96 to 0.99 places the poles closer to the unit circle, making the notch/resonant transition narrower and the transient decay longer, while retaining causal stability because  $r < 1$ .

### Solution 9

A conjugate pole pair generates damped sinusoidal impulse-response terms proportional to  $r^n \cos(n\theta + \phi)$  for a causal system. Thus  $r$  controls the envelope. As  $r \rightarrow 1^-$  the decay becomes slower. Geometrically, the unit-circle evaluation point near angle  $\theta$  gets closer to the pole, shrinking a denominator distance and producing a larger, narrower response peak. The time-domain and frequency-domain observations are two views of the same pole geometry.

## Weekly Quiz — Week 3

Suggested time: 15 minutes. Total: 20 points.

### Problem Q1: ROC and DTFT (4 pts)

$X(z)=1/(1-1.25z^{-1})$  with ROC  $|z|<1.25$ . Does the DTFT exist? Is the corresponding sequence right-sided or left-sided?

### Problem Q2: Inverse z-transform (4 pts)

Find  $x[n]$  for  $X(z)=1/(1-0.6z^{-1})$ , ROC  $|z|>0.6$ .

### Problem Q3: Transfer function (5 pts)

For  $y[n]-0.5y[n-1]=x[n]-x[n-1]$ , find  $H(z)$ , its pole, and its zero.

### Problem Q4: Causal + stable rule (3 pts)

A causal rational system has a pole at  $z=1.02$ . Can it be BIBO stable? Briefly justify.

### Problem Q5: Geometry (4 pts)

What happens to  $|H(\exp(j\omega))|$  at a frequency whose unit-circle point coincides exactly with a zero? What happens qualitatively near a pole that lies very close to the unit circle?

## Weekly Quiz Solutions and Grading Guide

### Solution Q1

Yes, because the unit circle  $|z|=1$  is contained in  $|z|<1.25$ . The sequence is left-sided:  $x[n]=-(1.25)^n u[-n-1]$ . Award 2 pts for DTFT/ROC reasoning and 2 pts for support/sequence interpretation.

### Solution Q2

$x[n]=(0.6)^n u[n]$ . Award 2 pts for the transform pair, 1 pt for right-sided support, and 1 pt for using the ROC correctly.

### Solution Q3

$H(z)=(1-z^{-1})/(1-0.5z^{-1})$ . Pole  $z=0.5$ ; zero  $z=1$ . Award 3 pts for  $H(z)$ , 1 pt pole, 1 pt zero.

### Solution Q4

No. Causality requires the ROC outside the outermost pole, hence  $|z|>1.02$ , which excludes the unit circle. Equivalently, a causal stable rational system must have all poles strictly inside  $|z|=1$ . Award 1 pt answer, 2 pts justification.

### Solution Q5

At a zero on the unit circle, a numerator distance is zero, so  $|H|=0$  (perfect notch). Near a pole close to the unit circle, a denominator distance is small, so  $|H|$  becomes large (resonant peak). Award 2 pts each.

**Instructor note:** For seniors, emphasize correct calculations and ROC interpretation. For graduate students, require explicit convergence/support justification and geometric reasoning.